USB Device Upgrade(onebin)

1. Overview

1.1. Upgrade Method

1. USB device upgrade in blank flash mode
2. USB device upgrade in Uboot mode
3. USB device upgrade in Kernel mode(uvc mode)
4. USB device upgrade in Kernel mode(USB mode)

1.2. Test Environment

pcb: ssd212 demo board

Kernel config: pioneer3_spinand_defconfig

project config: dispcam_p3_spinand.glibc-9.1.0-s01a.64.qfn128.demo_defconfig

usbtool: usbdownloadtool 2.1

2. USB Device Upgrade in Blank Flash Mode

- Scene

  It is suitable for scenario where there are complete pcb and usb interfaces, but nandflash or norflash is empty.

- Principle

  a. Execute rom code before powering on IC.

  b. Jump to nandflash/norflash to perform IPL according to the external HW config.

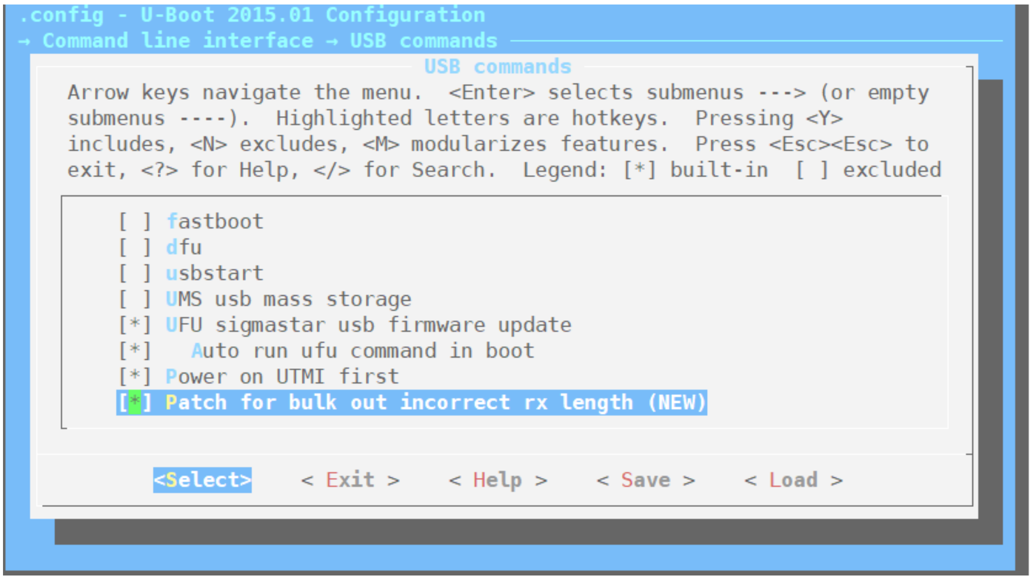
  c. Without IPL, the system will automatically enter the ufu upgrade mode. Pre-load `u-boot.bin` through pc tool.

- Requirement

  The pre-loaded `u-boot.bin` has ufu upgrade function.

- Step

  a. uboot config

  Rename the generated `u-boot_spinand.img.bin` or `u-boot.img.bin` to `u-boot.bin` (only for blank boot). This file will be used in the following steps.

  b. Compile SDK to generate an image upgrade file.

  c. Download the toolkit. (take nandflash as an example)

Use different toolkits to build different flash. (the corresponding SDK contains the upgrade tool)

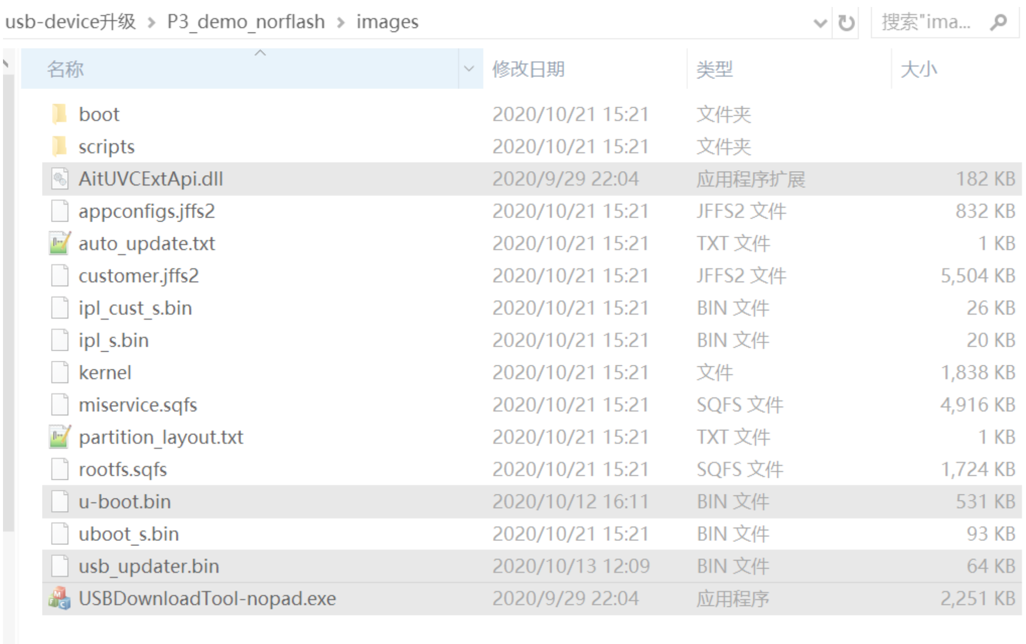
SPINOR FLASH upgrade kit:

```
usbloadertool_onebin_spinor_20210104.rar
```

SPINAND FLASH upgrade kit:

```
usbloadertool_onebin_spinand_20210104.rar.rar
```

Unzip the corresponding toolkit to the image path.

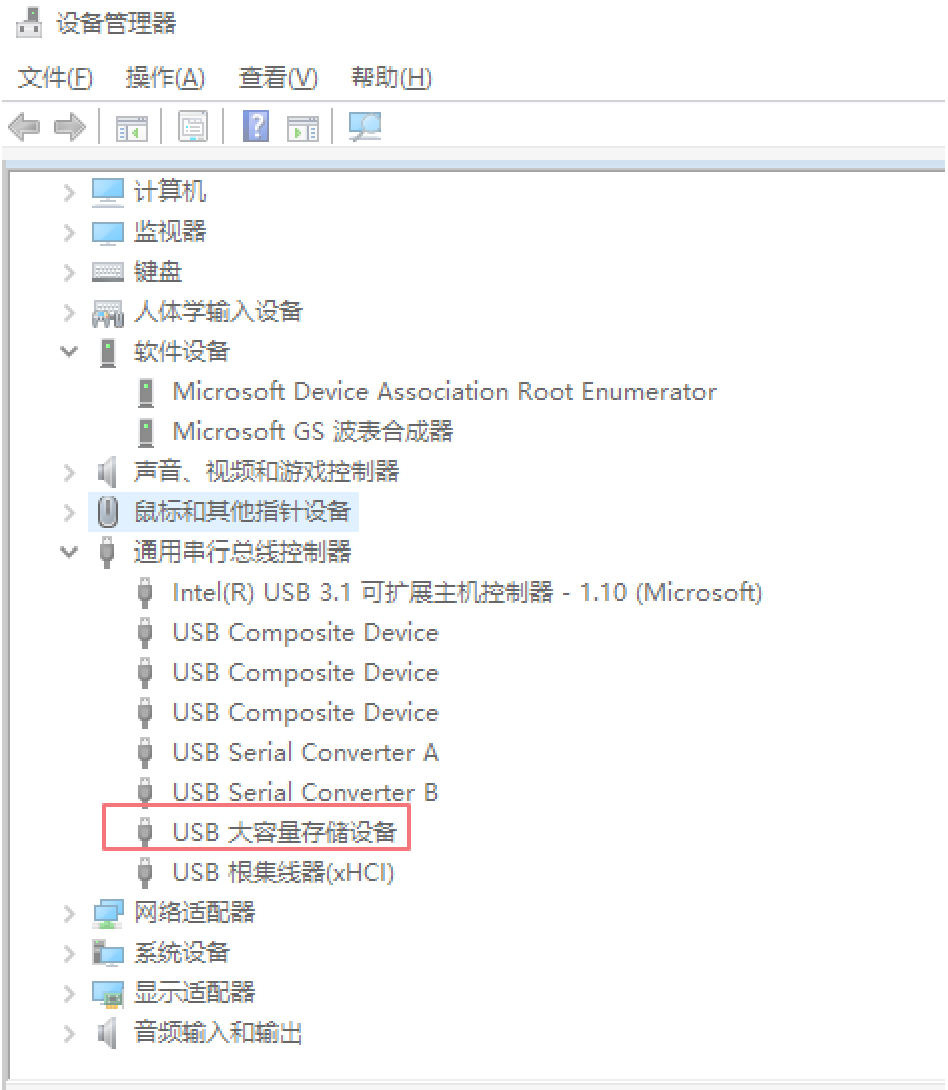


usb-device升级 > P3_demo_norflash > images

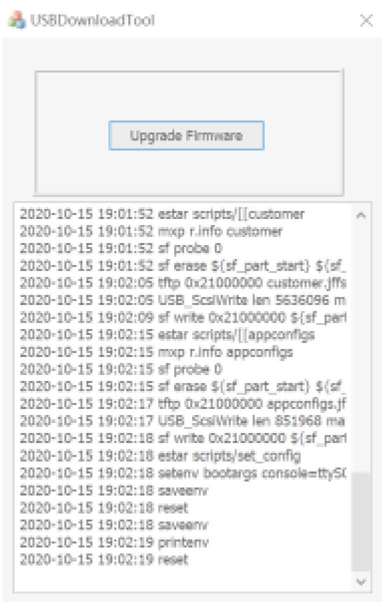
名称	修改日期	类型	大小
boot	2020/10/21 15:21	文件夹	
scripts	2020/10/21 15:21	文件夹	
AitUVCExtApi.dll	2020/9/29 22:04	应用程序扩展	182 KB
appconfigs.jffs2	2020/10/21 15:21	JFFS2 文件	832 KB
auto_update.txt	2020/10/21 15:21	TXT 文件	1 KB
customer.jffs2	2020/10/21 15:21	JFFS2 文件	5,504 KB
ipl_cust_s.bin	2020/10/21 15:21	BIN 文件	26 KB
ipl_s.bin	2020/10/21 15:21	BIN 文件	20 KB
kernel	2020/10/21 15:21	文件	1,838 KB
miservice.sqfs	2020/10/21 15:21	SQFS 文件	4,916 KB
partition_layout.txt	2020/10/21 15:21	TXT 文件	1 KB
rootfs.sqfs	2020/10/21 15:21	SQFS 文件	1,724 KB
u-boot.bin	2020/10/12 16:11	BIN 文件	531 KB
uboot_s.bin	2020/10/21 15:21	BIN 文件	93 KB
usb_updater.bin	2020/10/13 12:09	BIN 文件	64 KB
USBDownloadTool-nopad.exe	2020/9/29 22:04	应用程序	2,251 KB

d. Replace the generated `u-boot.bin` here.

e. Power on in empty flash mode and connect to usb. There is an access storage device on PC.



f. Execute `USBDownloadTool-nopad.exe` .



After the system is upgraded normally, the boot process can be viewed through the serial port.

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3. USB Device Upgrade in Uboot Mode

- Scene

  It is suitable for scenario where there are complete pcb and usb interfaces to ensure that the system can enter uboot mode

- Principle

  There is an uboot in norflash or nandfash, which supports PC upgrade in device mode. Complete the download and upgrade by executing uboot and pc software to interact.

- Requirement

  uboot has ufu upgrade function.

- Step

  a. uboot config

     Support ufu upgrade, but disabled automatic operation.

  b. Burn uboot with ufu upgrade function into flash.

     To use the USB device upgrade function in uboot, you need to set the following parameters in the boot mode, and then you can enter the device upgrade mode after booting.

     ```
     setenv ota_upgrade_status 1

     saveenv
     ```

  c. Download the toolkit. (take nandflash as an example)

     Use different toolkits to build different flash. (the corresponding SDK contains the upgrade tool)

     SPINOR FLASH upgrade kit:

```
usbloadertool_onebin_spinor_20210104.rar
```

SPINAND FLASH upgrade kit:

```
usbloadertool_onebin_spinand_20210104.rar.rar
```

Unzip the corresponding toolkit to the image path.

d. Reboot and enter the system, the display is as follows.

```
U-Boot 2015.01 (Oct 27 2020 - 16:49:36)

Version: P3gc2d7157
I2C:   ready
DRAM:
WARNING: Caches not enabled
MMC:   MStar SD/MMC: 0
nor_flash_mxp allocated success!!
Flash is detected (0x0709, 0xEF, 0x40, 0x18)
SF: Detected nor0 with total size 16 MiB
MXP found at mxp_offset[3]=0x00020000, size=0x1000
env_offset=0x6F000 env_size=0x1000
Flash is detected (0x0709, 0xEF, 0x40, 0x18)
SF: Detected nor0 with total size 16 MiB
In:    serial
Out:   serial
Err:   serial
Net:   MAC Address 00:30:1B:BA:02:DB
Auto-Negotiation...
AN failLink Status Speed:10 Full-duplex:0
Status Error!
sstar_emac
<USB> config miu select [70] [ef] [ef] [ef].
<USB>[UDC] PULL DOWN D+.
<USB>[UDC] PULL UP D+.
registered gadget driver 'msb250x_udc'
<USB>[LINK] High speed device.
<USB>[1][EN] bulk in with maxpacket/fifo(512/8192)
<USB>[2][EN] bulk out with maxpacket/fifo(512/1024)
dev->driver->setup failed. (-22)
```

e. Execute `USBDownloadTool-nopad.exe` .

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4. USB Device Upgrade in Kernel Mode (UVC Mode)

- Part

  Kernel config and APP

- Principle

a. Configure the uvc mode on the board.

b. Use pc tool to issue commands to the APP on the board.

c. Set the environment variables.

d. Use the ufu upgrade function in uboot to complete the upgrade.

To verify the usb upgrade in the kernel, first make sure that the usb device can be upgraded normally in uboot.

- Requirement

  0: uboot supports ufu upgrade. Refer to USB device upgrade in Uboot mode [#3-USB-Device-Upgrade-in-Uboot-Mode]

  1: Kernel configure uvc mode

  2: Need mi_uvc application

  3: USBDownloadTool upgraded to V2.1.

4.1. Kernel Config

There are two ways to increase the uvc config on the basis of the existing config:

1. Modify the default config of the current platform directly. (recommend)

   Add the following config:

```
CONFIG_MEDIA_SUPPORT=m
CONFIG_MEDIA_CAMERA_SUPPORT=y
CONFIG_MEDIA_CONTROLLER=y
CONFIG_VIDEO_DEV=m
CONFIG_VIDEO_V4L2=m
CONFIG_VIDEOBUF2_CORE=m
CONFIG_VIDEOBUF2_MEMOPS=m
CONFIG_VIDEOBUF2_VMALLOC=m
CONFIG_MEDIA_SUBDRV_AUTOSELECT=y
CONFIG_USB_GADGET=m
CONFIG_USB_GADGET_VBUS_DRAW=2
CONFIG_USB_GADGET_STORAGE_NUM_BUFFERS=2
CONFIG_USB_GADGET_SSTAR_DEVICE=m
CONFIG_USB_AVOID_SHORT_PACKET_IN_BULK_OUT_WITH_DMA_FOR_ETHERNET=y

CONFIG_USB_LIBCOMPOSITE=m
CONFIG_SS_GADGET=m
```

```
CONFIG_USB_F_UVC=m
CONFIG_USB_G_WEBCAM=m
CONFIG_USB_WEBCAM_UVC=y
CONFIG_MULTI_STREAM_FUNC_NUM=1
```

2. Use the default config of the current platform

```
make pioneer3_xxxx_defconfig
```

Add: make menuconfig

a. media framework config

```
-> Device Drivers
    -> Multimedia support

-> Device Drivers
    -> Multimedia support
        -> Cameras/video grabbers support

-> Device Drivers
    -> Multimedia support
        -> Media Controller API
```

Output module: media.ko videodev.ko v4l2-common.ko

3. usb Gadget framework config

```
    -> Device Drivers
        -> USB support

    -> Device Drivers
        -> USB support
            -> USB Gadget Support

Output module: usb-common.ko udc-core.ko
```

a. udc driver config

HW ip module, which can be configured according to the specific situation.

```
-> Device Drivers
    -> USB support
```

```
        -> USB Gadget Support
            -> USB Peripheral Controller
                -> Sstar USB 2.0 Device Controller
```

Output module: udc-msb250x.ko

4. gadget webcam

```
     -> Device Drivers
         -> USB support
             -> USB Gadget Support
                 -> USB Gadget Drivers

     -> Device Drivers
         -> USB support
         -> USB Gadget Support
             -> USB Gadget Drivers
                 -> USB Webcam Gadget

     -> Device Drivers
         -> USB support
             -> USB Gadget Support
                 -> USB Peripheral Controller
                     -> Sstar USB 2.0 Device Controller
                         -> Avoid short packet in bulk out with
DMA for ethernet

Output module: libcomposite.ko、 videobuf2-core.ko、 videobuf2-
v4l2.ko、 videobuf2-memops.ko、 videobuf2-vmalloc.ko 、
usb_f_uvc.ko g_webcam.ko
```

Project config:

Because there is no UVC related config added, the following two files need to be manually modified when compiling uvc function:

1. If you use the default config in kernel config, you need to modify project/makefile, comment out `$(MAKE)linux-kernel` in the image target to avoid recompiling the kernel to overwrite the previously manually compiled one;
2. Modify project/kbuild/customize/4.9.84/p3/dispcam/kernel_mod_list_late

```
media.ko
videodev.ko
```

```
v4l2-common.ko
usb-common.ko
#usbcore.ko
#ehci-hcd.ko
#scsi_mod.ko
#usb-storage.ko
videobuf2-core.ko
videobuf2-v4l2.ko
videobuf2-memops.ko
videobuf2-vmalloc.ko
udc-core.ko
libcomposite.ko
usb_f_uvc.ko
udc-msb250x.ko
g_webcam.ko streaming_maxpacket=3072 streaming_maxburst=13
uac_function_enable=0
```

4.2. Application

Please refer to myuvc_remove_media.zip for source code.

Application compilation (omitted)

Execute mi_uvc

Code:

```
case UVC_SET_CUR:
    if (cfg->bCmdCap & CAP_SET_CUR_CMD) {
        UVC_INFO("/etc/fw_setenv ota_upgrade_status 1\n");
        system("/etc/fw_setenv ota_upgrade_status 1");
       system("sleep 1");
        system("reboot");
        if (cfg->bInfoCap & INFO_AUTO_MODE_SUPPORT) {

        }
    }
    else {
        goto invalid_req;
    }
```

Process:

Compile the kernel, then execute application after updating the image.

After connecting usb of the pc, execute USBDownloadTool.exe. When the board receives the cmd of uvc, it will reset env and automatically restart to enter the uboot upgrade mode.

Click "upgrade firmware" to enter the upgrade state.

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5. USB Device Upgrade in Kernel Mode (USB Mode)

- Scene

  When it is not in uvc mode. Such as USB mode.

- Principle

  By default, USBDownloadTool first detects whether the device is uvc, if it is, it uses uvc mode; if not, it uses the extended upgrade. To support extended upgrades, USBDownloadTool sends a setup command by EP0 to replace the XU command, and notifies the board to upgrade. (Note: The special setup command has been added to the usb gadget driver, update to the corresponding version to upgrade. `composite.c` has been updated)

- Requirement

  0: uboot supports ufu upgrade.

  1: USB mode is supported by Kernel

  2: USBDownloadTool upgraded to V2.1.

- Config

  a. uboot config, refer to USB Device Upgrade in Uboot Mode [#3-USB-Device-Upgrade-in-Uboot-Mode]

  b. usb device mass storage function is supported by kernel

- Driver config

- usb Gadget framework config

  ```
  -> Device Drivers
      -> USB support
  ```

```
-> Device Drivers
    -> USB support
        -> USB Gadget Support
```

Output module: usb-common.ko udc-core.ko

- Configure to support configfs

```
-> Device Drivers
    -> USB Gadget Support
        -> USB functions configurable through configfs
```

If configured as a module, you need to install the corresponding driver: libcomposite.ko

- ip related driver

Such as: USB device2.0 driver

```
-> Device Drivers
    -> USB support
    -> USB Gadget Support
        -> USB Peripheral Controller
        -> Sstar USB 2.0 Device Controller
```

Generated driver: udc-msb250x.ko

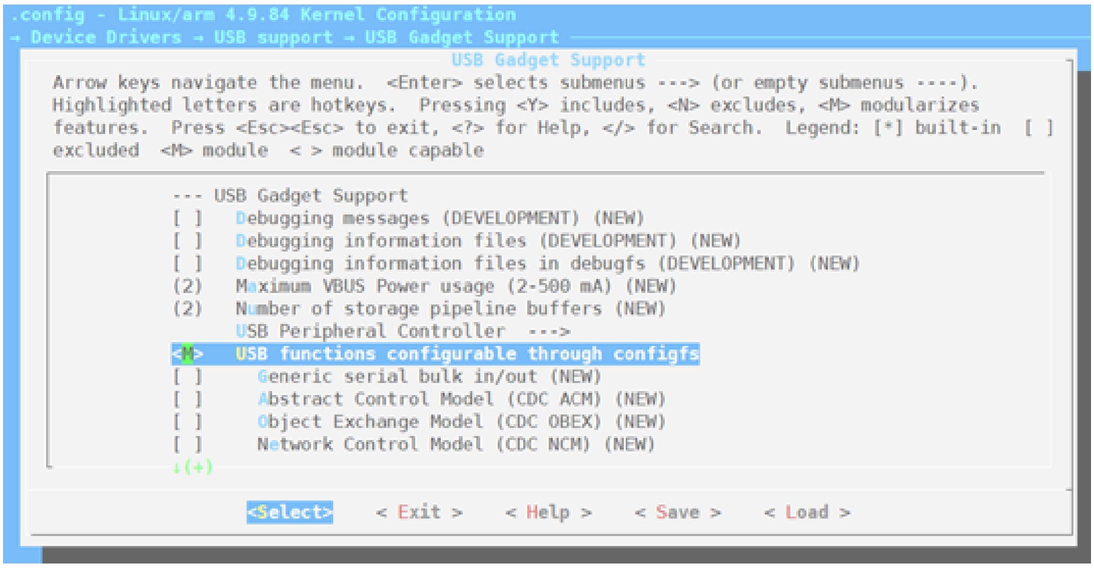
- usb function related driver

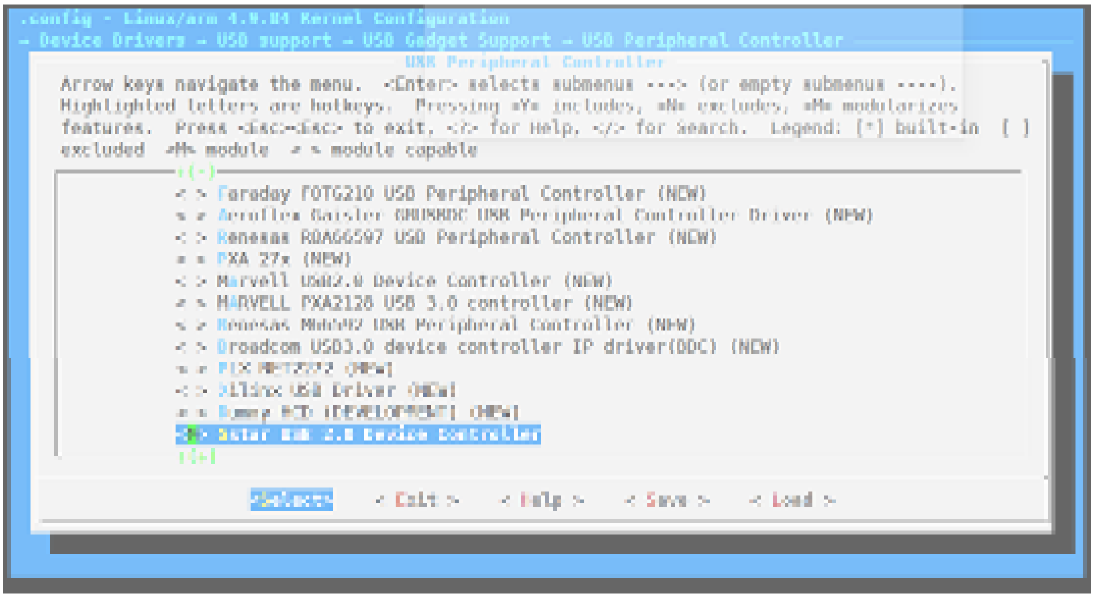
Configure sub-option: (need to deal with the dependency problem of function)

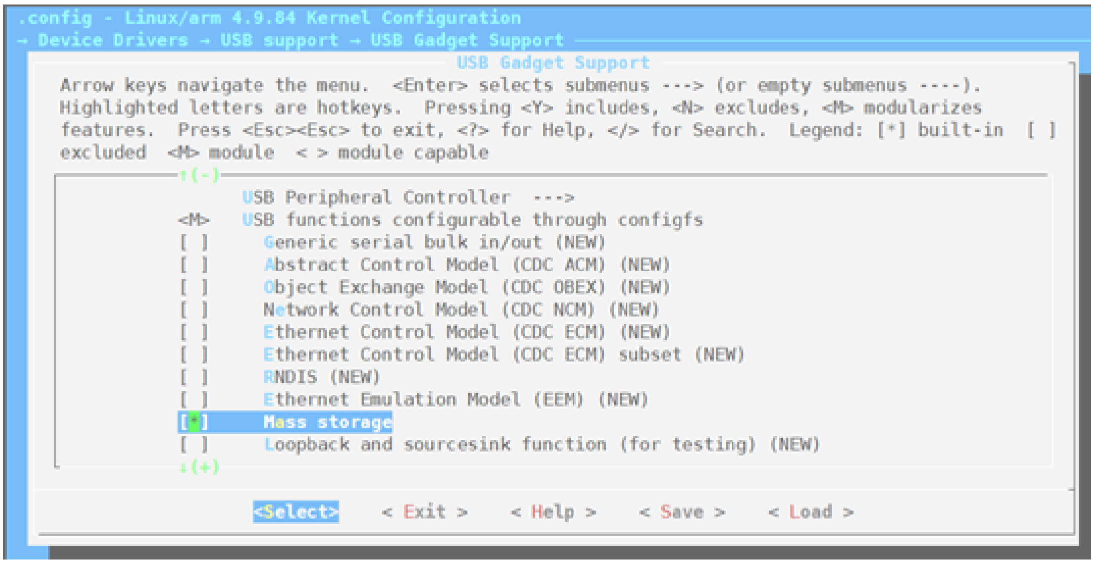
```
-> Device Drivers
    -> USB support
    -> USB Gadget Support
        -> USB functions configurable through configfs
```

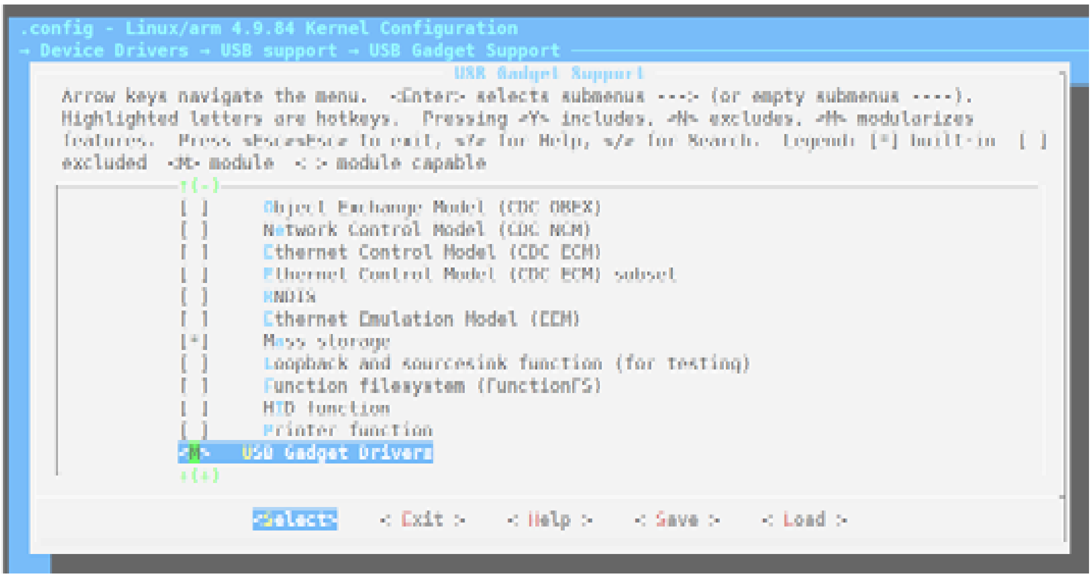
- mass storage

```
-> Device Drivers
    -> USB support -> USB Gadget Support -> USB Gadget Drivers
-> Mass Storage Gadget generated driver: usb_f_mass_storage.ko
```


.config - Linux/arm 4.9.84 Kernel Configuration
→ Device Drivers → USB support → USB Gadget Support
USB Gadget Support
Arrow keys navigate the menu. <Enter> selects submenus ---> (or empty submenus ----). Highlighted letters are hotkeys. Pressing <Y> includes, <N> excludes, <M> modularizes features. Press <Esc><Esc> to exit, <?> for Help, </> for Search. Legend: [*] built-in [ ] excluded <M> module < > module capable
--- USB Gadget Support
[ ] Debugging messages (DEVELOPMENT) (NEW)
[ ] Debugging information files (DEVELOPMENT) (NEW)
[ ] Debugging information files in debugfs (DEVELOPMENT) (NEW)
(2) Maximum VBUS Power usage (2-500 mA) (NEW)
(2) Number of storage pipeline buffers (NEW)
USB Peripheral Controller --->
<M> USB functions configurable through configfs
[ ] Generic serial bulk in/out (NEW)
[ ] Abstract Control Model (CDC ACM) (NEW)
[ ] Object Exchange Model (CDC OBEX) (NEW)
[ ] Network Control Model (CDC NCM) (NEW)
↓(+)
<Select> < Exit > < Help > < Save > < Load >


→ Device Drivers → USB support → USB Gadget Support → USB Peripheral Controller
Arrow keys navigate the menu. <Enter> selects submenus ---> (or empty submenus ----).
< > Faraday FOTG210 USB Peripheral Controller (NEW)
< > Renesas R8A66597 USB Peripheral Controller (NEW)
< > PXA 27x (NEW)
< > Marvell USB2.0 Device Controller (NEW)
< > MARVELL PXA2128 USB 3.0 controller (NEW)
< > Broadcom USB3.0 device controller IP driver(BDC) (NEW)


.config - Linux/arm 4.9.84 Kernel Configuration
→ Device Drivers → USB support → USB Gadget Support
USB Gadget Support
Arrow keys navigate the menu. <Enter> selects submenus ---> (or empty submenus ----). Highlighted letters are hotkeys. Pressing <Y> includes, <N> excludes, <M> modularizes features. Press <Esc><Esc> to exit, <?> for Help, </> for Search. Legend: [*] built-in [ ] excluded <M> module < > module capable
↑(-)
USB Peripheral Controller --->
<M> USB functions configurable through configfs
[ ] Generic serial bulk in/out (NEW)
[ ] Abstract Control Model (CDC ACM) (NEW)
[ ] Object Exchange Model (CDC OBEX) (NEW)
[ ] Network Control Model (CDC NCM) (NEW)
[ ] Ethernet Control Model (CDC ECM) (NEW)
[ ] Ethernet Control Model (CDC ECM) subset (NEW)
[ ] RNDIS (NEW)
[ ] Ethernet Emulation Model (EEM) (NEW)
[*] Mass storage
[ ] Loopback and sourcesink function (for testing) (NEW)
↓(+)
<Select> < Exit > < Help > < Save > < Load >



- insmod ko

  Modify project/kbuild/customize/4.9.84/p3/dispcam/kernel_mod_list_late:

  ```
  usb-common.ko
  udc-core.ko
  libcomposite.ko
  udc-msb250x.ko
  usb_f_mass_storage.ko
  ```

  Process:

  a. Update image

  b. Confirm to load the above ko files

  c. Execute script and configure USB parameters

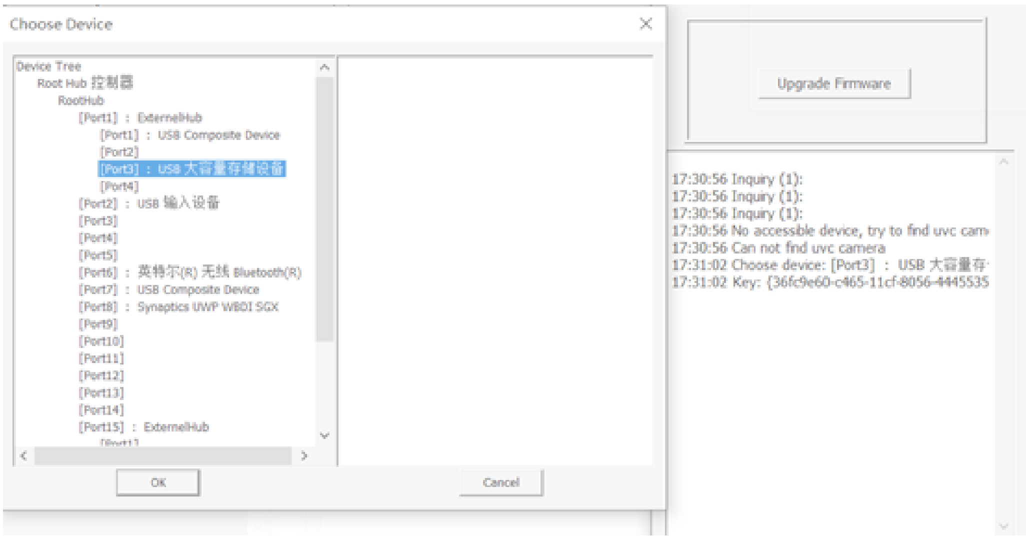
     Create file: dd if=/dev/zero of=/customer/disk.img bs=1M count=20

     ```
     ./usb_storage.sh
     ```

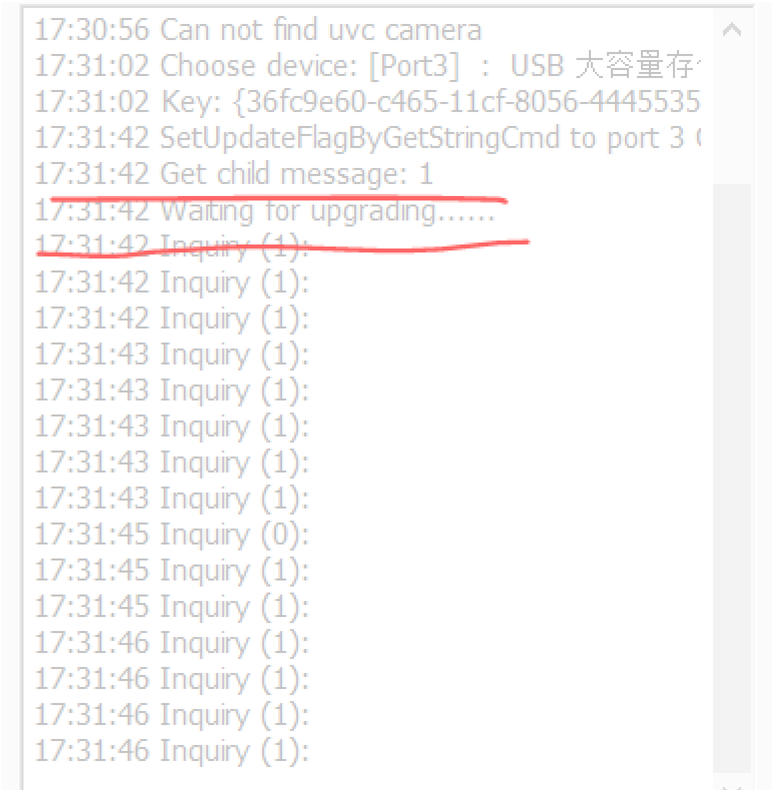
  d. Execute upgrade script

     ```
     ./fwupdated &
     ```

  e. Execute USBDownloadTool 2.1.exe



f. Select the storage device and click ok. The upgrade process will be shown in the figure, and it can also be observed by the serial port



g. Related configuration and image test reference images-usbotg-storage.7z [media/images-usbotg-storage.7z]

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6. Note

Note for unabled upgrade:

1. Make sure that the blank chip can also be upgraded normally, if not, check the hardware.
2. Make sure that the upgrade can be completed in uboot. If not, check the hardware and uboot config.
3. In the kernel mode, you can normally receive the upgrade command of the uvc tool (you can restart to enter the uboot upgrade mode). Make sure mi_uvc is executed
4. If more than one uvc device is detected on the pc, please disable the camera device.

Provide myuvc_remove_media.zip [media/myuvc_remove_media.zip] and images-usbotg-storage.7z [media/images-usbotg-storage.7z] for reference.